Lesson 17: Bounded Linear Operators and Dual Spaces

AI/ML Mathematics

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1 Linear Operators

Definition 1.1 (Linear Operator)**.** Let X and Y be vector spaces over the same field ($\mathbb{R}$ or $\mathbb{C}$). A map $T\colon X \to Y$ is a *linear operator* if for all $x, y \in X$ and all scalars α, β:

$$T(\alpha x + \beta y) = \alpha T(x) + \beta T(y).$$

Equivalently, T is additive ($T(x+y) = Tx + Ty$) and homogeneous ($T(\alpha x) = \alpha Tx$).

Remark 1.2. Every linear operator satisfies $T(0) = 0$ (set $\alpha = \beta = 0$). The *kernel* (or null space) is $\ker T = \{x \in X : Tx = 0\}$, and the *range* (or image) is $\operatorname{ran}(T) = \{Tx : x \in X\}$. Both are subspaces of X and Y respectively.

2 Bounded Linear Operators and the Operator Norm

Definition 2.1 (Bounded Linear Operator)**.** Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces. A linear operator $T\colon X \to Y$ is *bounded* if there exists a constant $M \geq 0$ such that

$$\|Tx\|_Y \leq M\|x\|_X \quad \text{for all } x \in X.$$

The infimum of all such constants M is the *operator norm* of T, denoted $\|T\|$.

Proposition 2.2 (Equivalent Formulations of the Operator Norm)**.** *For a bounded linear operator $T\colon X \to Y$, the following expressions are all equal:*

(i) $\|T\|_1 := \sup\limits_{\substack{x \in X \\ x \neq 0}} \dfrac{\|Tx\|_Y}{\|x\|_X}$,

(ii) $\|T\|_2 := \sup\limits_{\|x\|_X = 1} \|Tx\|_Y$,

(iii) $\|T\|_3 := \sup\limits_{\|x\|_X \leq 1} \|Tx\|_Y$.

Moreover, each equals $\inf\{M \geq 0 : \|Tx\|_Y \leq M\|x\|_X$ *for all* $x \in X\}$.

Proof. We show $\|T\|_1 = \|T\|_2 = \|T\|_3$.

$\|T\|_1 = \|T\|_2$**:** For any $x \neq 0$, set $u = x/\|x\|_X$, so $\|u\|_X = 1$ and $\|Tx\|_Y/\|x\|_X = \|T(x/\|x\|_X)\|_Y = \|Tu\|_Y$. Thus $\{\|Tx\|_Y/\|x\|_X : x \neq 0\} = \{\|Tu\|_Y : \|u\|_X = 1\}$, and their suprema are equal.

$\|T\|_2 \leq \|T\|_3$**:** The set $\{x : \|x\|_X = 1\}$ is contained in $\{x : \|x\|_X \leq 1\}$, so the supremum over the larger set is at least as large.

$\|T\|_3 \leq \|T\|_1$**:** For $x = 0$, $\|Tx\|_Y = 0 \leq \|T\|_1$. For $x \neq 0$ with $\|x\|_X \leq 1$:

$$\|Tx\|_Y = \|x\|_X \cdot \frac{\|Tx\|_Y}{\|x\|_X} \leq 1 \cdot \|T\|_1 = \|T\|_1.$$

Taking the supremum over $\|x\|_X \leq 1$: $\|T\|_3 \leq \|T\|_1$.

Combining: $\|T\|_1 = \|T\|_2 \leq \|T\|_3 \leq \|T\|_1$, so all three are equal. We denote the common value by $\|T\|$.

$\|T\| = \inf\{M : \|Tx\|_Y \leq M\|x\|_X \ \forall x\}$**:** Let M_0 denote this infimum. If $\|Tx\|_Y \leq M\|x\|_X$ for all x, then for $x \neq 0$, $\|Tx\|_Y/\|x\|_X \leq M$, so $\|T\|_1 \leq M$. Taking the infimum over such M: $\|T\| \leq M_0$. Conversely, for all x, $\|Tx\|_Y \leq \|T\|_1\|x\|_X$ (for $x \neq 0$ this follows from the definition of sup; for $x = 0$ it is trivial), so $\|T\|$ is one such M, giving $M_0 \leq \|T\|$. Thus $\|T\| = M_0$. $\square$

Remark 2.3. The operator norm satisfies $\|Tx\|_Y \leq \|T\| \cdot \|x\|_X$ for all $x \in X$. This is the defining property that makes $\|T\|$ useful: it converts an operator into a scalar bound.

3 Bounded if and only if Continuous

Theorem 3.1 (Bounded $\Leftrightarrow$ Continuous)**.** *Let $T\colon X \to Y$ be a linear operator between normed spaces. Then T is bounded if and only if T is continuous.*

Proof. **($\Rightarrow$) Bounded implies continuous.** Assume T is bounded with $\|Tx\|_Y \leq \|T\| \cdot \|x\|_X$ for all x. For any $x, y \in X$:

$$\|Tx - Ty\|_Y = \|T(x-y)\|_Y \leq \|T\| \cdot \|x-y\|_X.$$

This shows T is Lipschitz continuous with Lipschitz constant $\|T\|$. In particular, T is (uniformly) continuous.

($\Leftarrow$) Continuous implies bounded. We prove the contrapositive: if T is not bounded, then T is not continuous (in fact, not continuous at 0).

Suppose T is unbounded. Then for each $n \in \mathbb{N}$, there exists $x_n \in X$ with $\|x_n\|_X = 1$ and $\|Tx_n\|_Y > n$. (If no such x_n existed, $\|Tx\|_Y \leq n$ for all $\|x\|_X = 1$, contradicting unboundedness.)

Define $y_n = x_n/n$. Then:

$$\|y_n\|_X = \frac{\|x_n\|_X}{n} = \frac{1}{n} \to 0, \quad \text{so } y_n \to 0 \text{ in } X.$$

However:

$$\|Ty_n\|_Y = \frac{\|Tx_n\|_Y}{n} > \frac{n}{n} = 1 \quad \text{for all } n.$$

So $Ty_n \not\to 0 = T(0)$, which means T is not continuous at 0.

Since T is linear, continuity at 0 implies continuity everywhere (see the remark below), so T is not continuous at any point. □

Remark 3.2. For a linear operator T, continuity at a single point implies continuity everywhere. If T is continuous at x_0 and $x_n \to x$, then $x_n - x + x_0 \to x_0$, so $T(x_n - x + x_0) \to T(x_0)$. By linearity, $Tx_n - Tx + Tx_0 \to Tx_0$, hence $Tx_n \to Tx$. Thus T is continuous at x.

Bounded = Continuous for Linear Operators

For linear operators between normed spaces, the following are equivalent: (i) bounded, (ii) continuous, (iii) continuous at 0, (iv) continuous at some point. This equivalence is specific to linear operators – for general maps between metric spaces, boundedness and continuity are independent properties.

Unbounded Linear Operators Exist

Not every linear operator between normed spaces is bounded. Consider the space $P[0,1]$ of polynomials on $[0,1]$ with the sup norm, and the differentiation operator $D(f) = f'$. This is linear, but unbounded: $\|x^n\|_\infty = 1$ yet $\|D(x^n)\|_\infty = \|nx^{n-1}\|_\infty = n \to \infty$. Note that $P[0,1]$ is *not* a Banach space (it is not complete); the operator D cannot be defined on all of $C[0,1]$ as a bounded operator.

4 The Space $B(X, Y)$

Definition 4.1 (Space of Bounded Linear Operators)**.** For normed spaces X and Y, we denote by $B(X, Y)$ the set of all bounded linear operators $T\colon X \to Y$, equipped with the operator norm. When $Y = X$, we write $B(X) = B(X, X)$.

Theorem 4.2 ($B(X, Y)$ Is a Normed Space)**.** *For any normed spaces X and Y, the space $B(X, Y)$ with the operator norm is a normed space.*

Proof. $B(X,Y)$ **is a vector space.** For $T, S \in B(X,Y)$ and scalars α, β, the map $(\alpha T + \beta S)(x) = \alpha Tx + \beta Sx$ is linear (immediate from linearity of T and S) and bounded: $\|(\alpha T + \beta S)x\|_Y \leq |\alpha|\|Tx\|_Y + |\beta|\|Sx\|_Y \leq (|\alpha|\|T\| + |\beta|\|S\|)\|x\|_X$. The zero operator $0\colon x \mapsto 0$ is bounded with $\|0\| = 0$, and serves as the additive identity.

The operator norm satisfies the norm axioms.

?? (Positive definiteness):

$$\begin{aligned}\|T\| = 0 &\iff \sup_{\|x\|_X \leq 1} \|Tx\|_Y = 0 \\ &\iff \|Tx\|_Y = 0 \text{ for all } \|x\|_X \leq 1 \\ &\iff Tx = 0 \text{ for all } \|x\|_X \leq 1.\end{aligned}$$

But any $x \in X$ can be written as $\|x\|_X \cdot (x/\|x\|_X)$ for $x \neq 0$ (with $\|x/\|x\|_X\|_X = 1$), so $Tx = \|x\|_X \cdot T(x/\|x\|_X) = 0$ by homogeneity. Thus $\|T\| = 0 \iff T = 0$.

?? (Homogeneity): For scalar α,

$$\|\alpha T\| = \sup_{\|x\|_X \leq 1} \|\alpha Tx\|_Y = \sup_{\|x\|_X \leq 1} |\alpha|\|Tx\|_Y = |\alpha| \sup_{\|x\|_X \leq 1} \|Tx\|_Y = |\alpha|\|T\|.$$

?? (Triangle inequality):

$$\|T+S\| = \sup_{\|x\|_X \leq 1} \|(T+S)x\|_Y = \sup_{\|x\|_X \leq 1} \|Tx + Sx\|_Y \leq \sup_{\|x\|_X \leq 1} \big(\|Tx\|_Y + \|Sx\|_Y\big) \leq \|T\| + \|S\|.$$

The last inequality holds because $\|Tx\|_Y \leq \|T\|$ and $\|Sx\|_Y \leq \|S\|$ whenever $\|x\|_X \leq 1$. □

Remark 4.3 (Submultiplicativity). If $T \in B(Y,Z)$ and $S \in B(X,Y)$, then the composition $TS \in B(X,Z)$ satisfies $\|TS\| \leq \|T\| \cdot \|S\|$. Indeed: $\|TSx\|_Z \leq \|T\|\|Sx\|_Y \leq \|T\|\|S\|\|x\|_X$. In particular, $B(X)$ is a *Banach algebra* when X is Banach (an algebra with a submultiplicative norm that is complete).

Theorem 4.4 ($B(X,Y)$ Is Banach When Y Is Banach)**.** *If Y is a Banach space, then $B(X,Y)$ is a Banach space.*

Proof. Let $(T_n)_{n=1}^{\infty}$ be a Cauchy sequence in $B(X,Y)$. We must show that it converges to some $T \in B(X,Y)$.

Step 1: Define the pointwise limit. For each $x \in X$:

$$\|T_n x - T_m x\|_Y = \|(T_n - T_m)x\|_Y \leq \|T_n - T_m\| \cdot \|x\|_X \to 0 \quad \text{as } n, m \to \infty.$$

So $(T_n x)_{n=1}^{\infty}$ is a Cauchy sequence in Y. Since Y is Banach, it converges. Define $Tx = \lim_{n\to\infty} T_n x$ for each $x \in X$.

Step 2: T is linear. For $x, y \in X$ and scalars α, β:

$$\begin{aligned}T(\alpha x + \beta y) &= \lim_{n\to\infty} T_n(\alpha x + \beta y) \\ &= \lim_{n\to\infty} \big(\alpha T_n x + \beta T_n y\big) \\ &= \alpha \lim_{n\to\infty} T_n x + \beta \lim_{n\to\infty} T_n y \\ &= \alpha Tx + \beta Ty.\end{aligned}$$

The limit distributes by the continuity of vector space operations in Y.

Step 3: T is bounded. Since (T_n) is Cauchy in $B(X,Y)$, it is bounded: there exists $M > 0$ with $\|T_n\| \leq M$ for all n. For any $x \in X$:

$$\|Tx\|_Y = \lim_{n\to\infty} \|T_n x\|_Y \leq \lim_{n\to\infty} \|T_n\|\|x\|_X \leq M\|x\|_X.$$

(We used continuity of the norm: $y_n \to y$ implies $\|y_n\| \to \|y\|$.) Thus $T \in B(X,Y)$ with $\|T\| \le M$.

Step 4: $T_n \to T$ in operator norm. Given $\varepsilon > 0$, choose N such that $\|T_n - T_m\| < \varepsilon$ for all $n, m \ge N$. Fix $n \ge N$. For any x with $\|x\|_X \le 1$ and any $m \ge N$:

$$\|T_n x - T_m x\|_Y \le \|T_n - T_m\| \cdot \|x\|_X \le \varepsilon.$$

Let $m \to \infty$. By continuity of the norm in Y:

$$\|T_n x - Tx\|_Y = \lim_{m\to\infty} \|T_n x - T_m x\|_Y \le \varepsilon.$$

Taking the supremum over $\|x\|_X \le 1$:

$$\|T_n - T\| = \sup_{\|x\|_X \le 1} \|T_n x - Tx\|_Y \le \varepsilon \quad \text{for all } n \ge N.$$

Hence $T_n \to T$ in $B(X,Y)$. □

$B(X,Y)$ Inherits Completeness from Y

If Y is a Banach space, then the space of bounded linear operators $B(X,Y)$ is a Banach space. The completeness of Y is used in a single crucial place: to guarantee that the pointwise limits $Tx = \lim T_n x$ exist. Without this, the limit operator cannot even be defined.

5 Operators on Finite-Dimensional Spaces

Proposition 5.1 (All Linear Operators on Finite-Dimensional Spaces Are Bounded)**.** *If X is a finite-dimensional normed space and Y is any normed space, then every linear operator $T\colon X \to Y$ is bounded.*

Proof. Let $\dim X = n$ and let $\{e_1, \dots, e_n\}$ be a basis for X. Every $x \in X$ can be written as $x = \sum_{i=1}^n \alpha_i e_i$ for unique scalars α_i. The map $x \mapsto \sum_{i=1}^n |\alpha_i|$ defines a norm on X (call it $\|\cdot\|_1'$). By Theorem **??** from the previous lesson (all norms on $\mathbb{R}^n$ are equivalent), there exists $c > 0$ such that

$$\sum_{i=1}^n |\alpha_i| = \|x\|_1' \le c\,\|x\|_X \quad \text{for all } x \in X.$$

(The equivalence is applied after identifying X with $\mathbb{R}^n$ via the coordinate map $x \mapsto (\alpha_1, \dots, \alpha_n)$.)

Now estimate:

$$\|Tx\|_Y = \Big\|T\Big(\sum_{i=1}^n \alpha_i e_i\Big)\Big\|_Y = \Big\|\sum_{i=1}^n \alpha_i T e_i\Big\|_Y \le \sum_{i=1}^n |\alpha_i|\,\|Te_i\|_Y \le c\,\|x\|_X \cdot \max_{1\le i\le n} \|Te_i\|_Y.$$

So T is bounded with $\|T\| \le c \cdot \max_i \|Te_i\|_Y$. □

Remark 5.2. The converse is false: there exist bounded linear operators on infinite-dimensional spaces. The proposition says that in finite dimensions, *every* linear operator is bounded, so there is no distinction between "linear" and "bounded linear" in finite dimensions.

6 Important Examples

Example 6.1 (Matrix Operators)**.** Every $m \times n$ real matrix A defines a linear operator $T_A \colon \mathbb{R}^n \to \mathbb{R}^m$ by $T_A(x) = Ax$. Since $\mathbb{R}^n$ is finite-dimensional, T_A is bounded by Proposition 5.1. The operator norm depends on the choice of norms on $\mathbb{R}^n$ and $\mathbb{R}^m$:

- $\|T_A\|_{\infty\to\infty} = \max_{1\le i\le m} \sum_{j=1}^n |a_{ij}|$ (maximum absolute row sum),
- $\|T_A\|_{1\to 1} = \max_{1\le j\le n} \sum_{i=1}^m |a_{ij}|$ (maximum absolute column sum),
- $\|T_A\|_{2\to 2} = \sigma_{\max}(A)$ (largest singular value).

Proof of the $\|\cdot\|_{\infty\to\infty}$ formula. For $x \in \mathbb{R}^n$ with $\|x\|_\infty \le 1$:

$$\|Ax\|_\infty = \max_{1\le i\le m} \Big|\sum_{j=1}^n a_{ij}x_j\Big| \le \max_{1\le i\le m} \sum_{j=1}^n |a_{ij}||x_j| \le \max_{1\le i\le m} \sum_{j=1}^n |a_{ij}|.$$

For the reverse inequality, let i_0 achieve $\max_i \sum_j |a_{ij}|$. Define $x_j^* = \operatorname{sgn}(a_{i_0 j})$ for each j. Then $\|x^*\|_\infty = 1$ and $(Ax^*)_{i_0} = \sum_j a_{i_0 j} \operatorname{sgn}(a_{i_0 j}) = \sum_j |a_{i_0 j}|$. Thus $\|Ax^*\|_\infty \ge \sum_j |a_{i_0 j}| = \max_i \sum_j |a_{ij}|$. □

Example 6.2 (Integral Operator)**.** Let $k \in C([a,b] \times [a,b])$ be a continuous kernel function. Define $T \colon C[a,b] \to C[a,b]$ by

$$(Tf)(s) = \int_a^b k(s,t)\, f(t)\, dt.$$

T **is linear:** $(T(\alpha f + \beta g))(s) = \int_a^b k(s,t)[\alpha f(t) + \beta g(t)]\, dt = \alpha(Tf)(s) + \beta(Tg)(s)$.
T **is bounded:** For $f \in C[a,b]$,

$$|(Tf)(s)| = \Big|\int_a^b k(s,t) f(t)\, dt\Big| \le \int_a^b |k(s,t)|\, |f(t)|\, dt \le \|f\|_\infty \int_a^b |k(s,t)|\, dt.$$

Taking the maximum over s: $\|Tf\|_\infty \le \|f\|_\infty \cdot \max_{s\in[a,b]} \int_a^b |k(s,t)|\, dt$. So $\|T\| \le \max_s \int_a^b |k(s,t)|\, dt$.

Tf **is continuous:** For $s_1, s_2 \in [a,b]$, $|(Tf)(s_1) - (Tf)(s_2)| \le \|f\|_\infty \int_a^b |k(s_1,t) - k(s_2,t)|\, dt \to 0$ as $s_1 \to s_2$ by uniform continuity of k on the compact set $[a,b]^2$. So $Tf \in C[a,b]$.

Example 6.3 (Right Shift Operator on ℓ^p)**.** On ℓ^p $(1 \le p \le \infty)$, define the *right shift* $R \colon \ell^p \to \ell^p$ by

$$R(x_1, x_2, x_3, \ldots) = (0, x_1, x_2, x_3, \ldots).$$

Linearity is clear.

Boundedness: $\|Rx\|_p^p = \sum_{n=2}^\infty |x_{n-1}|^p = \sum_{n=1}^\infty |x_n|^p = \|x\|_p^p$ for $1 \le p < \infty$, so $\|Rx\|_p = \|x\|_p$, giving $\|R\| = 1$. (Similarly $\|R\| = 1$ for $p = \infty$.)

The right shift is an isometry (norm-preserving). It is injective but not surjective: $(1, 0, 0, \ldots)$ is not in the range of R.

Example 6.4 (Left Shift Operator on ℓ^p)**.** The *left shift* $L \colon \ell^p \to \ell^p$ defined by $L(x_1, x_2, x_3, \ldots) = (x_2, x_3, x_4, \ldots)$ is bounded with $\|L\| = 1$. Indeed, $\|Lx\|_p^p = \sum_{n=1}^\infty |x_{n+1}|^p \le \sum_{n=1}^\infty |x_n|^p = \|x\|_p^p$, and equality is achieved by $x = (0, 1, 0, 0, \ldots)$. The left shift is surjective but not injective: $L(1, 0, 0, \ldots) = (0, 0, \ldots)$.

7 Dual Spaces

Definition 7.1 (Dual Space)**.** The *(continuous) dual space* of a normed space X is

$$X^* = B(X, \mathbb{R})$$

(or $B(X, \mathbb{C})$ if X is a complex space): the space of all bounded linear functionals on X, equipped with the operator norm

$$\|f\|_{X^*} = \sup_{\|x\|_X \le 1} |f(x)|.$$

Elements of X^* are called *bounded linear functionals* or *continuous linear functionals.*

Corollary 7.2 (The Dual Space Is Always Banach)**.** *For any normed space X, the dual space X^* is a Banach space.*

Proof. The scalar field $\mathbb{R}$ (or $\mathbb{C}$) is a Banach space (it is complete). By Theorem 4.4, $X^* = B(X, \mathbb{R})$ is a Banach space. □

Remark 7.3. This is a remarkable fact: the dual space X^* is always complete, *even if X itself is not.* For example, if X is the space of polynomials on $[0, 1]$ with the sup norm (which is not complete), its dual X^* is nonetheless a Banach space.

7.1 The Duality Pairing

Definition 7.4 (Duality Pairing)**.** For $f \in X^*$ and $x \in X$, we write $\langle f, x\rangle = f(x)$ and call this the *duality pairing* between X^* and X. It is bilinear: linear in f for fixed x and linear in x for fixed f.

Remark 7.5. The duality pairing satisfies $|\langle f, x\rangle| \le \|f\|_{X^*}\|x\|_X$, which is just the definition of the operator norm applied to the functional f.

7.2 Identification of Dual Spaces

Example 7.6 (Dual of ℓ^p for $1 \le p < \infty$)**.** For $1 \le p < \infty$ with conjugate exponent q defined by $1/p + 1/q = 1$ (with $q = \infty$ when $p = 1$), the dual space $(\ell^p)^*$ is isometrically isomorphic to ℓ^q. Concretely:

Every $y \in \ell^q$ defines a bounded functional on ℓ^p: Given $y = (y_n) \in \ell^q$, define $\varphi_y \colon \ell^p \to \mathbb{R}$ by

$$\varphi_y(x) = \sum_{n=1}^{\infty} y_n x_n.$$

This is well-defined by Hölder's inequality: $\sum |y_n x_n| \le \|y\|_q \|x\|_p < \infty$. Linearity is clear, and $\|\varphi_y\|_{(\ell^p)^*} \le \|y\|_q$.

The norm is achieved: Define $x^* \in \ell^p$ by $x^*_n = |y_n|^{q-1} \operatorname{sgn}(y_n)$ (so $|x^*_n|^p = |y_n|^{(q-1)p} = |y_n|^q$ since $(q-1)p = q$). Then $\|x^*\|_p^p = \sum |y_n|^q = \|y\|_q^q$, so $\|x^*\|_p = \|y\|_q^{q/p}$, and

$$\varphi_y(x^*) = \sum_n y_n |y_n|^{q-1} \operatorname{sgn}(y_n) = \sum_n |y_n|^q = \|y\|_q^q.$$

Thus $|\varphi_y(x^*)|/\|x^*\|_p = \|y\|_q^q / \|y\|_q^{q/p} = \|y\|_q^{q-q/p} = \|y\|_q$ (using $q - q/p = q(1 - 1/p) = q/q = 1$). Hence $\|\varphi_y\| = \|y\|_q$.

Every bounded functional has this form: Let $f \in (\ell^p)^*$. Define $y_n = f(e_n)$ where e_n is the n-th standard unit vector. For $1 < p < \infty$, one shows $(y_n) \in \ell^q$ using the bound $\sum_{n=1}^N |y_n|^q \le \|f\|^q$ (obtained by evaluating f on a suitably chosen x), and then $f = \varphi_y$ by density of finitely supported sequences.

The map $y \mapsto \varphi_y$ is therefore an isometric isomorphism $\ell^q \cong (\ell^p)^*$.

Example 7.7 (Dual of $\mathbb{R}^n$)**.** Consider $\mathbb{R}^n$ with the ℓ^2 norm. Every bounded linear functional $f \colon \mathbb{R}^n \to \mathbb{R}$ has the form $f(x) = \langle a, x\rangle = \sum_{i=1}^n a_i x_i$ for a unique $a \in \mathbb{R}^n$, with $\|f\|_{(\mathbb{R}^n)^*} = \|a\|_2$ (by Cauchy–Schwarz). Thus $(\mathbb{R}^n)^* \cong \mathbb{R}^n$ isometrically. More generally, $(\mathbb{R}^n, \|\cdot\|_p)^* \cong (\mathbb{R}^n, \|\cdot\|_q)$ with $1/p + 1/q = 1$.

Example 7.8 (Evaluation Functionals on $C[a,b]$)**.** For each $t_0 \in [a,b]$, the evaluation functional $\delta_{t_0} \colon C[a,b] \to \mathbb{R}$ defined by $\delta_{t_0}(f) = f(t_0)$ is a bounded linear functional with $\|\delta_{t_0}\| = 1$. Indeed, $|\delta_{t_0}(f)| = |f(t_0)| \le \|f\|_\infty$, so $\|\delta_{t_0}\| \le 1$. Equality is achieved by the constant function $f \equiv 1$: $|\delta_{t_0}(1)| = 1 = \|1\|_\infty$.

Example 7.9 (Integration as a Bounded Functional)**.** On $(C[a,b], \|\cdot\|_\infty)$, the integration functional $I(f) = \int_a^b f(t)\,dt$ is bounded linear with $\|I\| = b - a$. Indeed, $|I(f)| = |\int_a^b f(t)\,dt| \le \int_a^b |f(t)|\,dt \le (b-a)\|f\|_\infty$, so $\|I\| \le b - a$. The constant function $f \equiv 1$ gives $|I(1)| = b - a = (b-a)\|1\|_\infty$, so $\|I\| = b - a$.

Dual Spaces in Optimization and RL

In constrained optimization (relevant to constrained MDPs), Lagrange multipliers live in the dual of the constraint space. The KKT conditions state that at optimality, there exist dual variables $\lambda \in X^*$ such that the gradient of the Lagrangian vanishes. In RL, the dual of the state space $L^\infty(\mathcal{S})$ contains the space of signed measures, and the stationary distribution d^π of a policy π can be viewed as a dual object. The fact that dual spaces are always Banach (Corollary 7.2) ensures that these dual optimization problems are well-posed.

8 Exercises

Exercise 8.1 (Operator Norm of a Diagonal Matrix)**.** Let $D = \operatorname{diag}(d_1, \ldots, d_n)$ be a diagonal matrix. Compute the operator norm $\|D\|_{p\to p}$ for $p = 1, 2, \infty$.

Hint: All three answers involve $\max_i |d_i|$ for $p = 2$ and $p = \infty$. For $p = 1$, use the column-sum formula.

Exercise 8.2 (Composition of Bounded Operators)**.** Let X, Y, Z be normed spaces, $T \in B(X,Y)$ and $S \in B(Y,Z)$. Prove that $ST \in B(X,Z)$ and $\|ST\| \le \|S\| \cdot \|T\|$. Give an example showing that strict inequality can occur.

Hint: For the example, consider projection operators on $\mathbb{R}^2$.

Exercise 8.3 (Continuity of Operator Inversion)**.** Let X be a Banach space and $T \in B(X)$ with $\|T\| < 1$. Prove that $I - T$ is invertible in $B(X)$ and $(I-T)^{-1} = \sum_{n=0}^\infty T^n$ (the *Neumann series*), where $T^0 = I$ and $T^n = T \circ T^{n-1}$. Show that $\|(I-T)^{-1}\| \le 1/(1 - \|T\|)$.

Hint: Use the series characterization of Banach spaces (Theorem **??** from the previous lesson) to show that $\sum T^n$ converges in $B(X)$, then verify it is the inverse of $I - T$.

Exercise 8.4 (Dual of c_0)**.** Recall that $c_0 = \{x \in \ell^\infty : \lim_n x_n = 0\}$ with the sup norm. Prove that $(c_0)^* \cong \ell^1$ isometrically, i.e., every bounded linear functional f on c_0 has the form $f(x) = \sum_{n=1}^\infty y_n x_n$ for a unique $y \in \ell^1$, with $\|f\| = \|y\|_1$.

Hint: Define $y_n = f(e_n)$. Show $\sum |y_n| \le \|f\|$ by evaluating f on $x^{(N)} = \sum_{n=1}^N \operatorname{sgn}(y_n) e_n$.

Exercise 8.5 (Unbounded Operator on an Incomplete Space)**.** On the space $X = \{f \in C^1[0,1] : f(0) = 0\}$ with the sup norm $\|f\|_\infty$, show that the differentiation operator $D \colon X \to C[0,1]$ defined by $Df = f'$ is linear and unbounded.

Hint: Consider $f_n(t) = t^n/n$.

Exercise 8.6 (Isometries and Operator Norm)**.** Let $T \colon X \to Y$ be a linear operator between normed spaces.

(a) Show that if T is an isometry ($\|Tx\|_Y = \|x\|_X$ for all x), then $\|T\| = 1$ (provided $X \neq \{0\}$).

(b) Show that $\|T\| = 1$ does not imply T is an isometry. Give a concrete counterexample.

Hint: For (b), consider the left shift on ℓ^2.

References